AMINO-MAP

Step-by-step tutorial

2026

This is a step-by-step tutorial on how to use Amino-Map. The data used in this example are embedded in the .tsv file inside /public/data but can be replaced by the user using the base code that can be found at https://github.com/queraltmartinsaladich/amino-map

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Header and navigation panel

Re-directs to the README.md file that explains the base code structure and hierarchy of the repository for developers

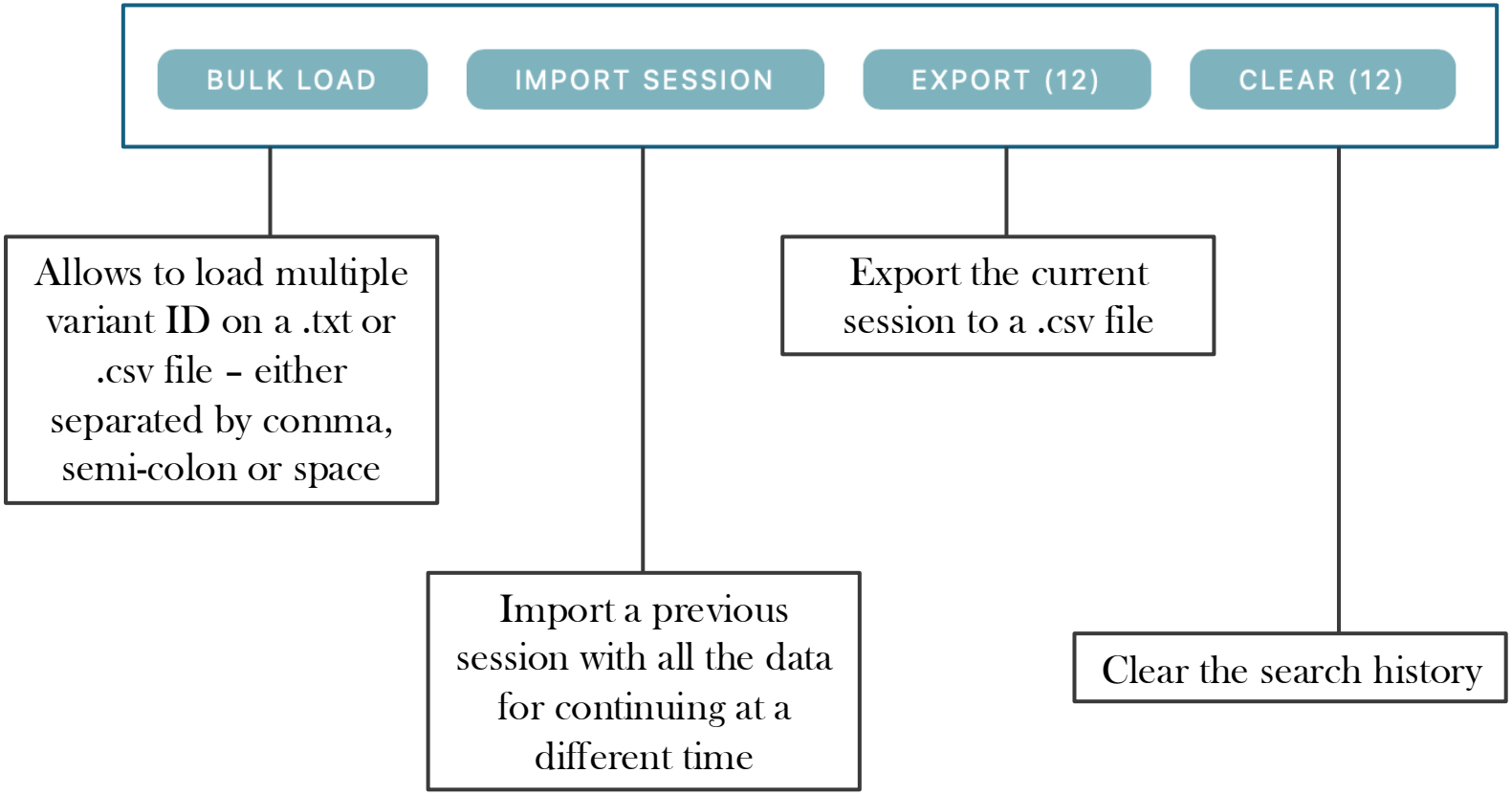
Re-directs to Beltrao Group website

Re-directs to this tutorial in PDF format

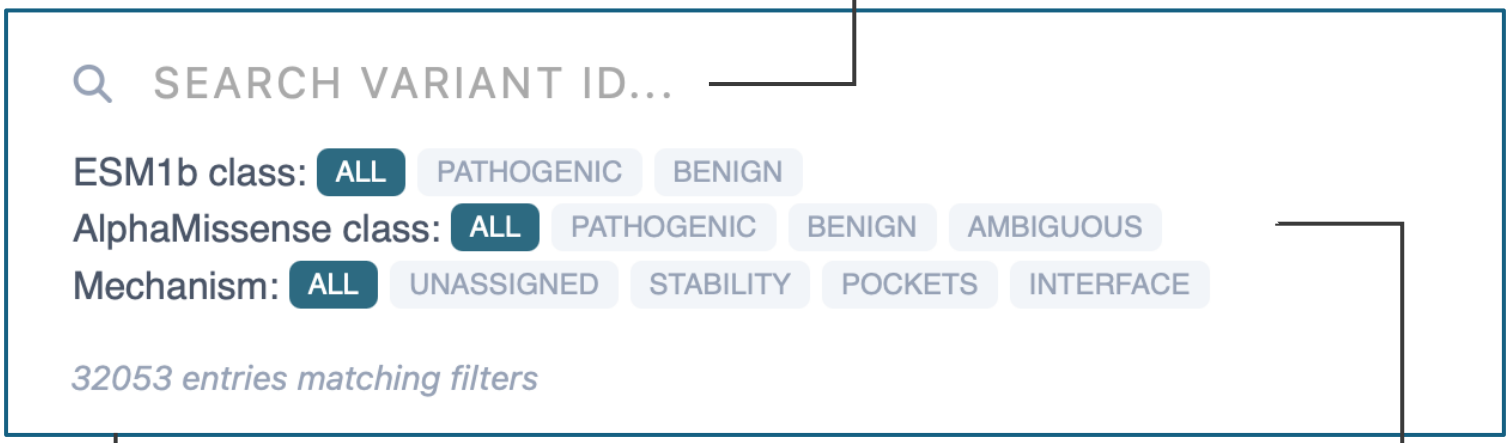
Re-directs to the repository for developers

Re-directs to the references.md file with the doi and links referenced in the app

Data management panel



Searching and filtering



Search table contents by ID

Visible entries

Filters. Active filters appear highlighted in ”azure” color and inactive are displayed with a lighter color

Table appearance

VARIANT ID	ESM1B CLASS	AM CLASS	MECHANISM
Q7Z4H8/A126C	PATHOGENIC	PATHOGENIC	UNASSIGNED
Q7Z4H8/A126D	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126E	PATHOGENIC	AMBIGUOUS	STABILITY
Q7Z4H8/A126F	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126G	PATHOGENIC	BENIGN	UNASSIGNED
Q7Z4H8/A126H	PATHOGENIC	PATHOGENIC	UNASSIGNED
Q7Z4H8/A126I	PATHOGENIC	PATHOGENIC	STABILITY
Q7Z4H8/A126K	PATHOGENIC	PATHOGENIC	UNASSIGNED
Q7Z4H8/A126L	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126M	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126N	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126P	PATHOGENIC	PATHOGENIC	STABILITY
Q7Z4H8/A126Q	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126R	PATHOGENIC	AMBIGUOUS	UNASSIGNED
Q7Z4H8/A126S	PATHOGENIC	BENIGN	UNASSIGNED
Q7Z4H8/A126T	PATHOGENIC	BENIGN	STABILITY
Q7Z4H8/A126V	PATHOGENIC	AMBIGUOUS	STABILITY
Q7Z4H8/A126W	PATHOGENIC	PATHOGENIC	UNASSIGNED
Q7Z4H8/A126Y	PATHOGENIC	PATHOGENIC	UNASSIGNED
Q7Z4H8/A149C	PATHOGENIC	AMBIGUOUS	UNASSIGNED

PAGE 1 OF 1603

Selected variant for analysis

Highlighted when the mouse is hovering over the item

Move forwards/backwards on the table

Main analysis

Download as PDF or print the report for this variant

Close the report

ANALYSIS OF

Q7Z4H8/A126C

PROTEIN O-GLUCOSYLTRANSFERASE 3 {ECO:0000305|PUBMED:30127001}

GENE: POGLUT3 | ORGANISM: HOMO SAPIENS (HUMAN)

Protein glucosyltransferase that catalyzes the transfer of glucose from udp-glucose to a serine residue within the consensus sequence peptide c-x-n-t-x-g-s-f-x-c (pubmed:30127001). can also catalyze t...

View full UniProt.org entry

POSITION	126
SUBSTITUTION	A → C
PROPERTY SHIFT	Small hydrophobic → Sulfur-containing (nucleophilic)
CHARGE SHIFT	No charge shift
LOCAL GEOMETRY	Loop / Unstructured
STRUCTURAL FLEXIBILITY	Moderate / Enzymatic Hinge

Analysis for Q7Z4H8/A126C: information from www.uniprot.org

Protein ID Mutation at position 126, “A” has been replaced for “C”

Characteristics

CHARACTERISTICS

PATHOGENIC LABELS

PATHOGENIC ESM1b model

PATHOGENIC AlphaMissense model

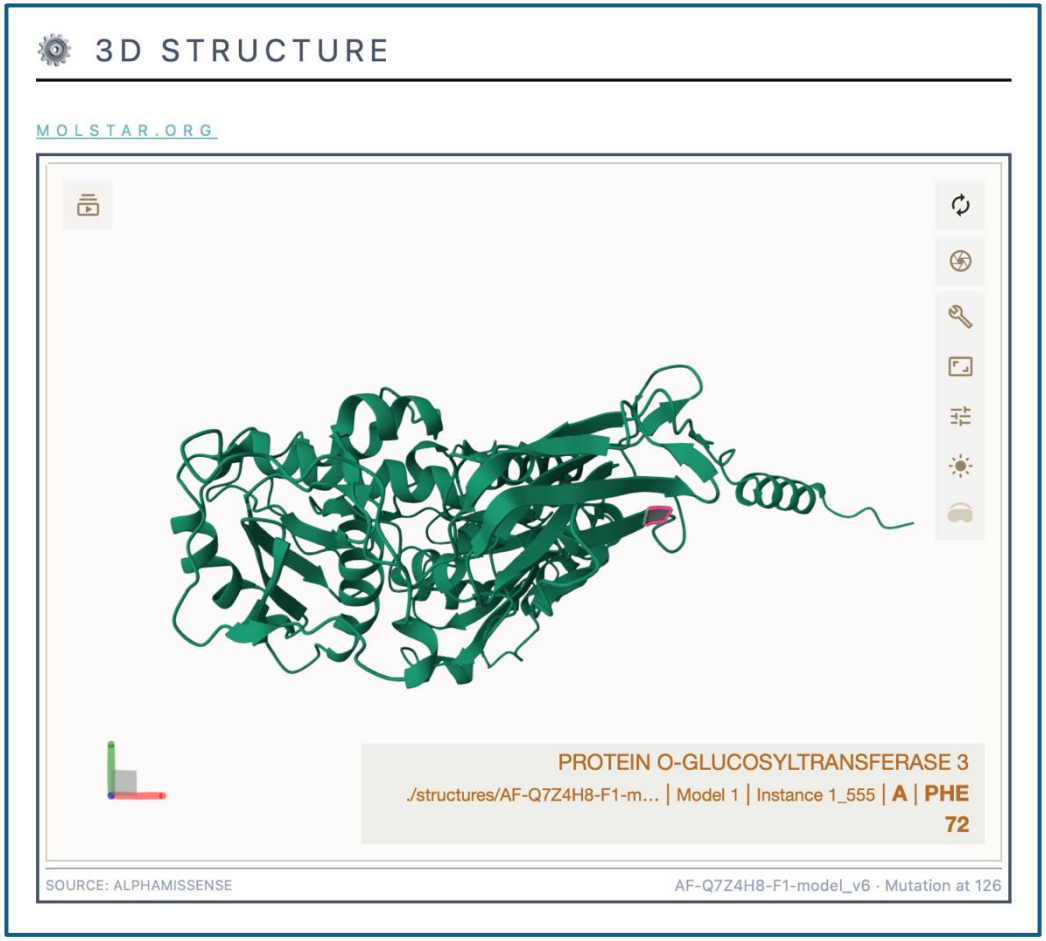
ESM1b likelihood ratio	-10.388
AM pathogenicity score	0.6158
Stability (ΔΔG)	-0.353881

SCORES

Destabilizing?	False
Interface score?	Unspecified
Interface?	False
Protein pockets?	False
Mechanism	Unassigned

Characteristics for Q7Z4H8/A126C: information from the table

3D structure



3D structure rendering using www.molstar.org, allows to select the AA and get the position

Sequencing

SEQUENCING

UNIPROT.ORG

```
  1 MRRLPRALLLQLRLALLVAAGAPEVLVSAPRSLVWGPGLQAAVVLPVRYF
 51 YLQAVNSEGQNLTRSPAGETPFKVVVKSLSPKELVRIHVPKPLDRNDGTF
101 LMRYRMYETVDEGLKIEVLYGDEHVAQSPYILKGPVYHEYCECPEDPQAW
151 QKTLSCPTKEPQIAKDFASFPSINLQQMLKEVPKRFGDERGAIVHYTILN
201 NHVYRRSLGKYTDFKMFSDEILLSLTRKVLLPDLEFYVNLGDWPLEHRKV
251 NGTPSPIPIISWCGSLDSRDVVLPTYDITHSMLEAMRGVTNDLLSIQGNT
301 GPSWINKTERAFFRGRDSREERLQLVQLSKENPQLLDAGITGYFFFQEKE
351 KELGKAKLMGFFDFFKYKYQVNVDGTVAAYRYPYLMLGDSLVLKQDSPYY
401 EHFYMALEPWKHYVPIKRNLSDLLEKVKWAKENDEEAKKIAKEGQLMARD
451 LLQPHRLYCYYYQVLQKYAERQSSKPEVRDGMELVPQPEDSTAICQCHRK
501 KPSREEL
```

SOURCE: UNIPROT Position: 126

Sequencing using www.uniprot.org highlights the AA at the mutation position

RNA expression

RNA EXPRESSION

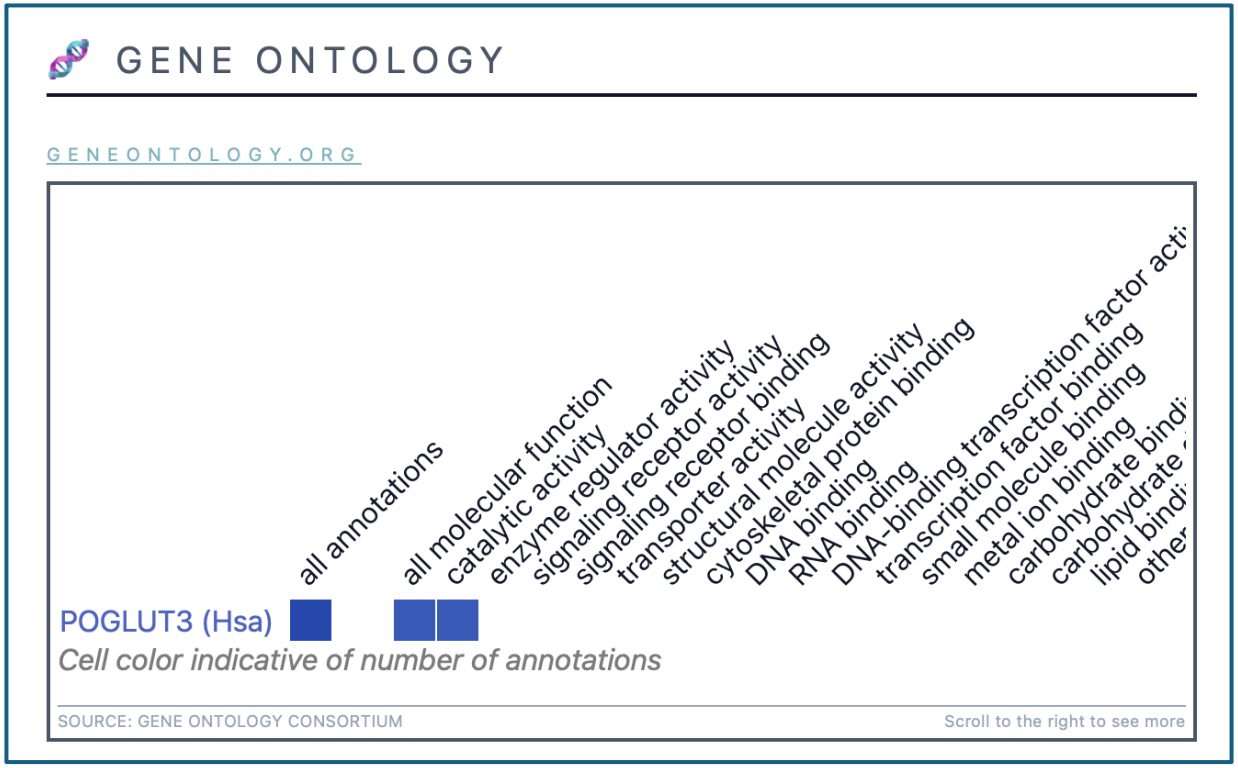
PROTEINATLAS.ORG

smooth muscle	34.7
ovary	34.0
endometrium 1	32.8
cervix	31.6
blood vessel	31.0
placenta	30.8
prostate	26.9
seminal vesicle	26.9
epididymis	26.4
parathyroid gland	23.9
adipose tissue	23.8
fallopian tube	22.1
urinary bladder	21.2
gallbladder	20.6
esophagus	19.9
vagina	18.2
appendix	16.4
colon	16.1
rectum	15.9
thymus	15.5

SOURCE: HUMAN PROTEIN ATLAS nTPM

Expression for different tissues according to www.proteinatlas.org

Gene ontology



Gene ontology ribbon according to www.geneontology.org

References

Original research paper

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Molstar

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> Webiste: https://molstar.org

UniProt

Bairoch, A., Apweiler, R., Wu, C. H., Barker, W. C., Boeckmann, B., Ferro, S., Gasteiger, E., Huang, H., Lopez, R., Magrane, M., Martin, M. J., Natale, D. A., O'Donovan, C., Redaschi, N., & Yeh, L. S. (2005). The Universal Protein Resource (UniProt). Nucleic acids research, 33(Database issue), D154–D159. https://doi.org/10.1093/nar/gki070

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> Webiste: https://alphamissense.hegelab.org

AlphaFold

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> Webiste: https://alphafold.ebi.ac.uk

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ESM1b

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> Repository: https://github.com/facebookresearch/esm